Legal Ethics Assistant: An Integrated Framework for Legal-Ethical Risk Analysis

1st Yanqing An
School of Artificial Intelligence and Data Science
University of Science and Technology of China
Hefei, China
anyq@mail.ustc.edu.cn

***Abstract*—With the growth of large language models and artificial intelligence, AI-based tools are increasingly being adopted. These technological advances bring convenience but also raise ethical concerns. For example, ordinary people may become embroiled in disputes or face unforeseen entrepreneurial risks. Non-professionals can obtain information about the potential consequences of an event from commercially available legal tools, while lawyers and other professionals can assess cases by searching case databases and generating similar-case reports. However, the results obtained in this way are often incomplete and drawn from multiple, fragmented sources. To provide a faster and more information-rich understanding of the potential consequences of an event, we propose the Legal Ethics Assistant. Specifically, we integrate lawyer competence assessment, legal judgment prediction, similar-case retrieval, and large language models to provide users with information relevant to the facts of a case, thereby assisting them in considering legal and ethical risks at the early stages of an entrepreneurial idea or a legal case. Our demo is available at https://bouncedude.github.io/test/demo.html.**

***Index Terms*—Legal intelligence, ethics, service**

I. Introduction

In recent years, with the rise of digital humans [1] and agentic reinforcement learning technologies, artificial intelligence has begun to be widely applied. While these advances bring convenience, they also raise ethical concerns. For example, ordinary people may become embroiled in disputes or face unforeseen entrepreneurial risks.

Numerous legal tools are available, including case retrieval databases such as China Judgments Online [1] and the Peking University Law Database [2], as well as plugins and mini-programs such as Faxiaofei [3] and Fabao AI [4].

Non-professionals often obtain information about the potential consequences of an event through the intelligent chat services provided by these applications. Lawyers and other professionals use case databases to generate similar-case search reports to assess cases, but the results obtained in this way are often partial and manually assembled. To provide users with a faster and more comprehensive understanding of the potential consequences of an event, we introduce the Legal Ethics Assistant. Specifically, we integrate lawyer competence assessment, legal judgment prediction, similar-case retrieval, and large language models to deliver case-related information to users.

With the rapid advancement of smart justice and inclusive legal services, ordinary citizens often face real pain points when encountering civil disputes: high barriers to legal information access, a lack of prediction of case outcomes, difficulty in quickly selecting suitable lawyers, and neglect of legal and ethical risks in cases. Traditional legal service channels suffer from high costs, slow response times, and information asymmetry. Existing legal AI tools mostly focus on legal provision retrieval and document generation, lacking integrated modeling of legal and ethical elements, and also lacking an integrated closed-loop capability encompassing "case analysis—risk identification—lawyer matching—precedent reference—outcome prediction." Moreover, model outputs often lack interpretability and can easily mislead users.

This demonstration system aims to build an intelligent auxiliary framework that integrates legal rules and legal ethical constraints. On the one hand, it helps lawyers and other professionals analyze cases quickly; on the other hand, it helps non-professionals recognize the potential legal risks of events. Its application scope is mainly civil dispute scenarios, including common livelihood cases such as private lending, family disputes, and small-amount contract disputes. The service targets include ordinary litigants, grassroots legal professionals, and community legal service personnel. Given natural language case fact text as input, the system automatically performs structured case analysis, ethical risk identification, cause of action determination, intelligent lawyer recommendation, similar-case retrieval, and judgment tendency prediction, while relying on a knowledge graph to ensure the interpretability of the reasoning process. Positioned as an auxiliary tool rather than a replacement for practicing lawyers in providing legal opinions, the system aims to lower the barriers to accessing legal services, promote technology for good, and support grassroots social governance.

In complex legal environments, ordinary people often face pain points such as difficulty in finding lawyers, difficulty in case assessment, and a lack of legal knowledge. Meanwhile, LegalTech must consider not only information matching but also *legal ethics* and *judicial fairness*. Traditional legal retrieval systems rely solely on keyword matching and lack a

[1]https://wenshu.court.gov.cn/
[2]https://www.pkulaw.com/
[3]https://sf.xfyun.cn/services/IFlyLegal
[4]https://www.fabao.law/

comprehensive understanding of the deeper logic of a case, the intentions of the parties involved, and the professional competence of lawyers.

The Legal Ethics Assistant aims to provide an end-to-end intelligent judicial service framework. Its core concept is "Tech for good," which drives social development and long-term stability by using advanced representation learning and user modeling technologies to transform unstructured case facts into structured legal insights, achieving a closed loop from "case input" to "precise lawyer recommendation," and then to "judgment prediction" and "similar-case retrieval."

This tool is divided into four parts: text input, semantic parsing and extraction, inference matching, and visual interactive output. The text input part allows users to describe case facts in natural language without requiring strictly formatted legal documents. The semantic parsing and information extraction part is based on a pre-trained language model and performs entity extraction (parties, assets, actions, temporal relationships) and relation extraction and identification, while simultaneously identifying implicit legal and ethical conflicts in the case and distinguishing between legal facts and ethical risks. The inference matching part first encodes massive amounts of publicly available judgment documents, learning vector representations of causes of action, points of contention, and judgment rules; second, lawyer representation learning constructs lawyer vectors based on practice areas, historical cases, winning characteristics, and ethical evaluation dimensions; then, relation representation learning learns the relationships among cases, legal provisions, precedents, and lawyers. Finally, it retrieves similar precedents based on vector similarity and combines precedent features to infer judgment tendencies. The visual interactive output part encapsulates structured results, lawyer information, radar charts, and precedent details into a front-end interactive page.

Existing large language model technology has already been extensively developed and applied; however, specialised domains require more specialised knowledge and the fulfilment of corresponding requirements. To this end, we have introduced legal intelligence. This assistant employs multiple legal intelligence technologies. Legal intelligence aims to alleviate the heavy and repetitive workload of legal professionals and provide reliable references for those unfamiliar with the legal field [2], including tasks such as judgment prediction, case retrieval, user recommendation, and legal summarization. First, we utilize legal judgment prediction technology, for example, in criminal [3], civil [4], and multi-defendant cases [5]. Second, to address the issue that large language models merely offer suggestions without recommending specific lawyers(e.g. Spark [5]), we use user modeling methods to create high-dimensional vector representations of lawyers, cases, and users [6]. By constructing complex relationships involving "lawyer representation" and "case representation," the system can capture implicit connections between entities and quantify lawyers' expertise in specific areas such as "Property Division," "Family," and "Gift Disputes," visualized through radar charts. Finally, to address the issue of large language models generating hallucinations and providing practitioners with fake cases(e.g. ChatGPT [7]), we use a case retrieval technology [8] to model the semi-structured text of legal documents and recommend similar cases. In addition, the Big Prophecy model is used for basic tasks, such as using NLP techniques to extract entities (plaintiff, defendant) and perform cause-of-action analysis from user-input natural language case information (e.g., disputes over gifts in extramarital affairs), as well as providing case summaries [9] and suggestions. There may well be more specialised large language model applications for entity extraction and summarisation techniques in the future. We hope that using legal intelligence can improve the interpretability of AI output and mitigate the legal risks associated with black-box prediction.

This assistant can analyze natural language case information and automatically identify parties and legal relationships; automatically determine the cause of action and provide legal and ethical risk warnings; offer multi-dimensional intelligent lawyer recommendations, including a radar chart of lawyer capabilities, representative cases, and professional background; retrieve similar prior cases and display details; and generate judgment tendency predictions, case summaries, and evidence collection suggestions. Its main functions are divided into three points. First, intelligent case analysis and lawyer recommendations. Users input case facts, and the system automatically identifies a case overview (parties, case type, cause of action) and provides a list of highly matched lawyers based on the underlying recommendation algorithm. Each lawyer is accompanied by a radar chart and precise tags.

Second, lawyer profiles, mainly presenting the lawyer's comprehensive data (years of practice, number of cases handled, and win rate), detailed radar chart capability profiles, and representative cases (case time, outcome, amount involved). Users can build deep trust in lawyers through this page and view their case-handling style.

Third, judgment prediction and similar-case retrieval. After analyzing the case, the system provides a prediction of the "Possible Judgment" and retrieves highly similar past cases. Users can click on a similar case (e.g., Zhang v. Li, a dispute over a gift contract) to seamlessly jump to a detailed "Case Details" page. This page breaks down the entire judicial process in the form of a case timeline, claims, reasoning (with specific legal provisions), and the final judgment.

Our contributions are as follows:

- We incorporate legal ethics elements into the closed loop of judicial intelligent modeling. Unlike most legal AIs that only focus on legal provisions and judgments, we simultaneously identify ethical risks during the case analysis stage, achieving joint reasoning based on legal rules and ethical constraints.
- Our joint representation learning framework models both the case and the lawyer simultaneously, enabling fine-grained case-lawyer matching rather than simply filtering lawyers by case type.

[5]https://xinghuo.xfyun.cn/

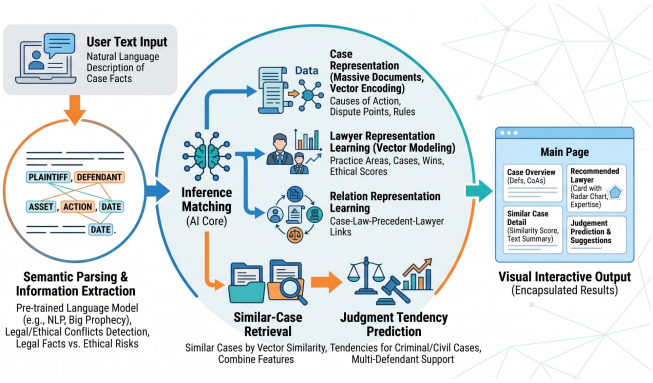


Fig. 1. Framework of Legal Ethics Assistant.

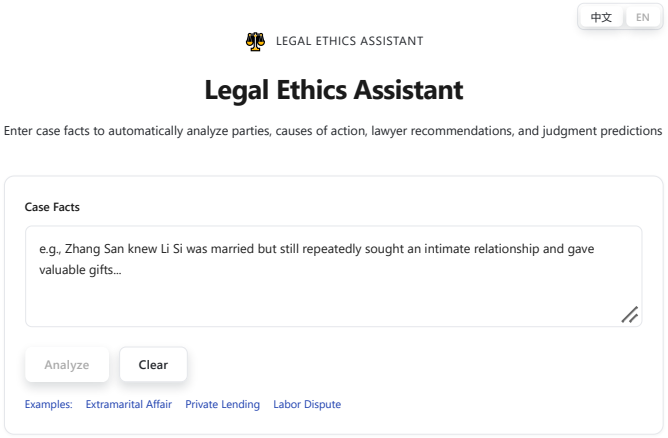


Fig. 2. Case fact input page of Legal Ethics Assistant.

- We design an integrated, lightweight interactive workflow for ordinary users, connecting the entire process of "inputting case details → risk identification → finding a lawyer → checking precedents → predicting future trends," rather than using fragmented and independent search tools.

II. System Overview

The framework of the system is shown as Figure 1. The Legal Ethics Assistant analyzes case facts to provide matching lawyers, similar cases, judgment tendencies, and recommendations. Figure 2 shows the initial fact input page. Figure 3 shows the lawyer detail page, which presents areas of expertise and a radar chart of competency scores. Figure 4 displays the main page, integrating the case overview (number of defendants, cause of action, case type), recommended lawyers, similar cases, judgment tendency, and suggestions; the number of defendants and party information are used for multi-defendant judgment prediction.

III. Conclusion

This paper proposes a Legal Ethics Assistant tool for both non-professional users and legal professionals. This tool utilizes a large amount of publicly available court judgment

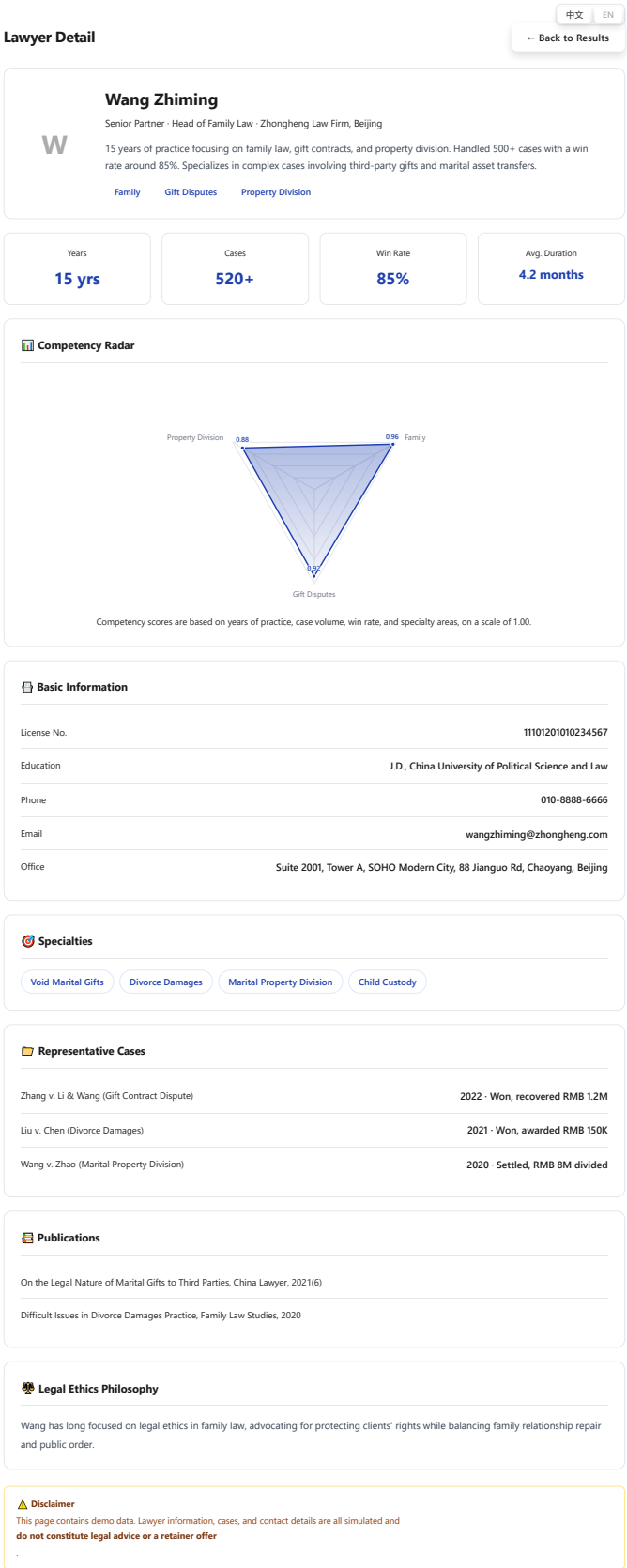


Fig. 3. Lawyer detail page of Legal Ethics Assistant.

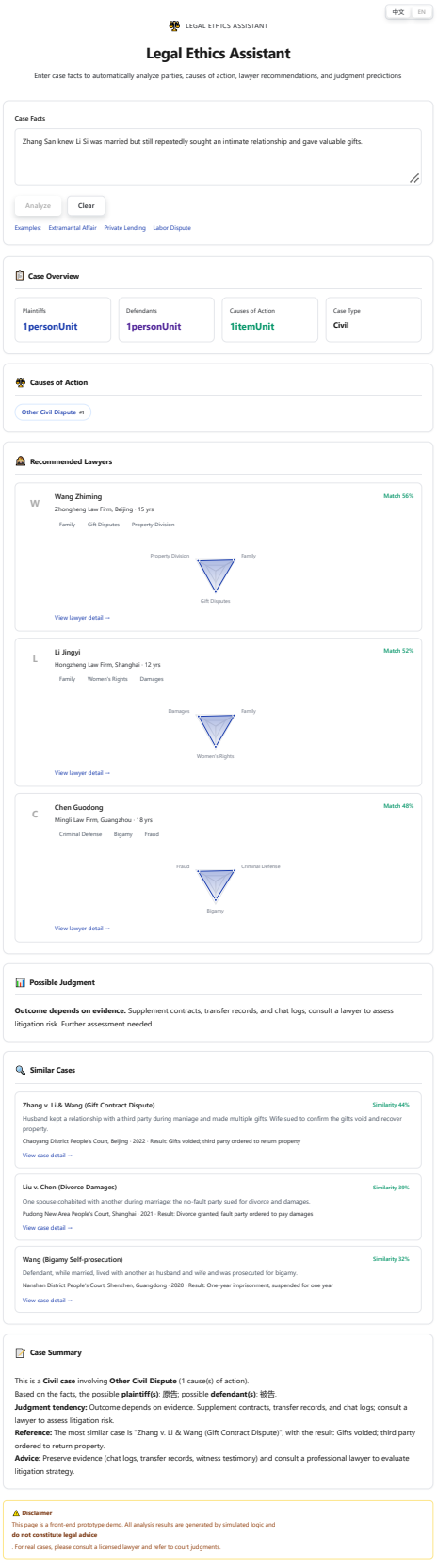


Fig. 4. Main page of Legal Ethics Assistant.

data online, recommending lawyers and similar cases based on case descriptions, and providing judgment tendencies and suggestions. This helps mitigate risks before or in the early stages of legal proceedings, bridging the information gap for those involved or legal professionals. By integrating lawyer and case information and continuously supplementing data and knowledge, the reliability and effectiveness of the recommendations are improved. Finally, by providing judgment tendencies and suggestions, the assistant guides users, realizing the positive impact of technology on society and maintaining long-term social stability.

The system is of significant novelty and research importance. At the practical level, the system reduces the cost of initial legal consultation for the public, helps parties identify case risks in advance, rationally anticipate judgments, and reduce irrational rights protection; it also assists grassroots legal service personnel in improving work efficiency. At the technical level, this solution provides an implementation approach for explainable legal AI that integrates ethical constraints, knowledge graphs, and dual-representation learning, balancing AI's deductive capabilities with the rigor required by legal scenarios. At the value level, the project adheres to a "Tech for good" orientation, clearly defining the boundaries of AI-assisted tools, ensuring it does not replace practicing lawyers, and balancing the needs of inclusive legal services with legal security risks, providing valuable reference for AI-enabled grassroots social governance.

REFERENCES

[1] K. Zhang, W. Gong, Q. Liu, J. Lu, M. Bao, X. Liu, L. Zhu, and E. Chen, "Intelligent interaction platform for personalized digital tutor: Advancing empathetic and adaptive learning experiences," in *Companion Proceedings of the ACM on Web Conference 2025*, 2025, pp. 2947–2950.

[2] H. Zhong, C. Xiao, C. Tu, T. Zhang, Z. Liu, and M. Sun, "How does nlp benefit legal system: A summary of legal artificial intelligence," in *Proceedings of the 58th annual meeting of the association for computational linguistics*, 2020, pp. 5218–5230.

[3] L. Yue, Q. Liu, B. Jin, H. Wu, K. Zhang, Y. An, M. Cheng, B. Yin, and D. Wu, "Neurjudge: A circumstance-aware neural framework for legal judgment prediction," in *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval*, 2021, pp. 973–982.

[4] L. Zhao, L. Yue, Y. An, Y. Zhang, J. Yu, Q. Liu, and E. Chen, "Cpee: C ivil case judgment p rediction centering on the trial mode of e ssential e lements," in *Proceedings of the 31st ACM international conference on information & knowledge management*, 2022, pp. 2691–2700.

[5] W. Huang, Y. Feng, C. Li, H. Wu, J. Ge, and V. Ng, "Cmdl: A large-scale chinese multi-defendant legal judgment prediction dataset," in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 5895–5906.

[6] Y. An, Q. Liu, H. Wu, K. Zhang, L. Yue, M. Cheng, H. Zhao, and E. Chen, "Lawyerpan: A proficiency assessment network for trial lawyers," in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 2021, pp. 5–13.

[7] J. Achiam, "Gpt-4 technical report," *arXiv preprint arXiv:2303.08774*, 2023.

[8] H. Li, Q. Ai, J. Chen, Q. Dong, Y. Wu, Y. Liu, C. Chen, and Q. Tian, "Sailer: structure-aware pre-trained language model for legal case retrieval," in *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2023, pp. 1035–1044.

[9] P. Bhattacharya, K. Hiware, S. Rajgaria, N. Pochhi, K. Ghosh, and S. Ghosh, "A comparative study of summarization algorithms applied to legal case judgments," in *European conference on information retrieval*. Springer, 2019, pp. 413–428.